The Impact of the Paris Accords on CO2 Levels

MA578 - Bayesian Statistics Project

Alen Abdildayev

1. Introduction

The Paris Accords are an international treaty on climate change that were signed in 2016. Nearly every country is a signatory. The treaty's goal is to limit the rise in global temperatures by reducing greenhouse gas emissions as much as possible. Ultimately, greenhouse gas emissions should reach net zero by the middle of the 21st century. However, almost eight years have passed since the treaty was signed.

In this project, we ask whether carbon dioxide emissions have decreased noticeably in some countries. For each country, we estimate the difference in cumulative CO2 emissions over 2015-2024 between *(a)* a forecast produced by a model fit to pre-2015 data and *(b)* the actual observed emissions.

Because we use a relatively simple model, we cannot definitively attribute any observed changes in a country's CO2 emissions to the Paris Accords. Nevertheless, it is useful to identify countries that show noticeable changes. This question is interesting because it examines effectiveness at the country level rather than at the global level.

2. Dataset

Our data come from the *Our World in Data's CO2 and Greenhouse Gas Emissions* dataset (https://ourworldindata.org/co2-and-greenhouse-gas-emissions). The dataset provides long-term records of national and global greenhouse gas emissions. For this project we use only annual CO2 emissions by country.

A potential issue is that emissions in the dataset are estimates rather than direct measurements, which introduces uncertainty, especially for historical values. In addition, data may be missing for some countries (for example, countries that gained independence recently). Therefore, for forecasting we use observations from 1964 through 2015 (50 years).

3. Methods and Analysis

3.1 Sampling Model and Prior

Because a simple linear regression is insufficient for these data, we use the BSTS (Bayesian Structural Time Series) framework for this project.

Our sampling model is:

$$y_t = \mu_t + \epsilon_t, \epsilon_t \sim N(0, \sigma^2_{obs})$$

$$\mu_{t+1} = \mu_t + \beta_t + \eta_t, \eta_t \sim N(0, \sigma^2_{level})$$

$$\beta_{t+1} = \beta_t + \xi_t, \xi_t \sim N(0, \sigma^2_{slope})$$

y_t: observed carbon dioxide emissions in year t (million t).

μ_t: underlying "level" of emissions at year t.

β_t: slope of emissions at year t, which can change over time.

ϵ_t: observation noise (difference between observed y_t and true level μ_t).

η_t: innovation in the level (how much the level μ_t can change beyond the slope β_t).

ξ_t: innovation in the slope (how much the slope β_t can change year-to-year).

Our priors are:

$$\sigma_{level} \sim SdPrior(\sigma_{guess} = \alpha \cdot sd(y), n = 10, u = sd(y))$$
$$\sigma_{slope} \sim SdPrior(\sigma_{guess} = \alpha \cdot sd(y), n = 10, u = sd(y))$$

$sd(y)$: sample standard deviation of the observed series y_t for priors; we will use $\alpha = 0.1$ as prior;

σ_{level}: standard deviation of level innovations η_t; σ_{slope}: standard deviation of slope innovations ξ_t;

3.2 Inferential Approach

1. Bayesian target: We infer the joint posterior of the latent states μ_t,β_t and hyper-parameters σ_{level},σ_{slope},σ_{obs}) given observed CO2 data $y_{1:T}$:

$$p(\theta, s \mid y) \propto p(y \mid s, \theta) p(s \mid \theta) p(\theta)$$

2. Approximation: Then we use MCMC to draw samples $(\theta^{(m)}, s^{(m)})$,and then discarding burn-in (around 1/4 of total amount of samples). Posterior expectations are computed as averages over these draws.

3. Posterior predictive: Future CO2 y_{T+h} is predicted by simulating from each posterior draw.

4. Estimator / point forecast: Posterior mean of predictive draws is used as the point forecast.

5. Hypothesis test: For years 2016–2024, we compute the posterior distribution of total predicted minus observed emissions, Δ. Then we report $P(|\Delta| > 0)$ and a 95% credible interval; We reject Null Hypothesis if the interval excludes 0.

3.3 Sensitivity Analysis

Here we examine how sensitive the results are to the prior choice. As an example, we use the United States and compare three different values of α values: $0.05, 0.1$ and 0.2.

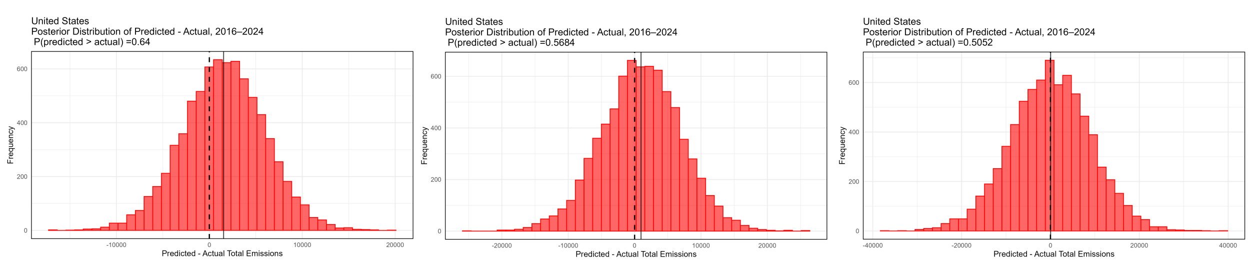

Figure 1: Posterior Distribution Difference of United States for: alpha=0.05,0.1 and 0.2

As the figure shows, the choice of prior has a noticeable impact on the posterior distribution of the total difference. This is expected, since intuitively, the prior tells us how large we believe state innovations can be on the order of $\alpha \cdot 100$ of the data's standard deviation, while still allowing flexibility and preventing unrealistically large jumps in the trend.

You can also check the residual diagnostics in the appendix for $\alpha = 0.1$, which indicate that the model fits well and the forecasts behave reasonably. With this in mind, we now turn to the global results.

4. Results

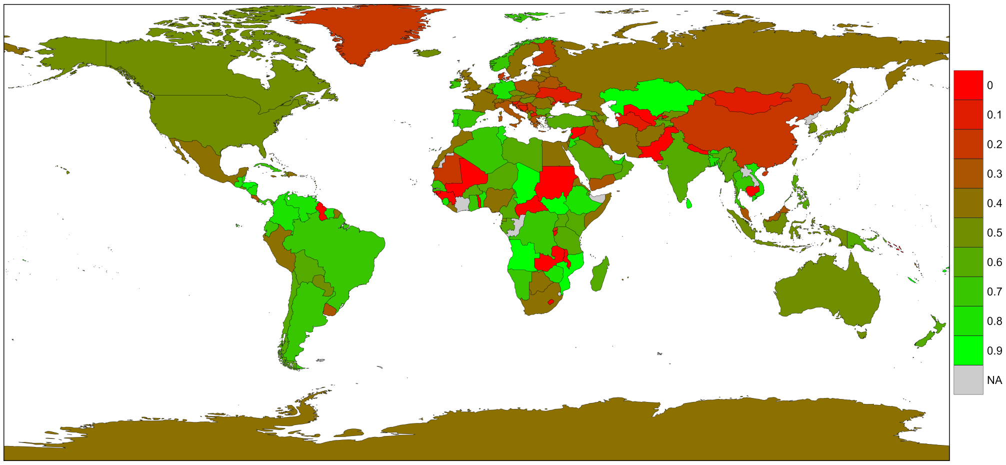

Figure 2: P[Predicted>Observed] by Country, 2016-2024, alpha=0.1

The figure above shows the posterior probability that the model's forecasted CO2 (based on pre-2015 data) exceeds the observed CO2 for 2016–2024. In other words, greener countries are those for which actual emissions tended to be lower than predicted; redder countries are those for which actual emissions were higher than expected.

Overall, most countries do not provide strong evidence against the null hypothesis: only a small number show 95% credible intervals that exclude zero.

Countries with lower amount of carbon emissions than predicted:

Country	P(predicted > actual)	CI Lower	CI Upper
South Sudan	0.99933333	4.567602	18.21746
Mozambique	0.99186667	6.430546	70.13750
Kazakhstan	0.97573333	3.895435	1854.439
Montserrat	0.97546667	0.002162896	0.5104619

Countries with higher amount of carbon emissions than predicted:

Country	P(predicted > actual)	CI Lower	CI Upper
Comoros	0.0002666667	-2.521903	-0.8323614
Nepal	0.0002666667	-84.92974	-26.33510
Burundi	0.0004000000	-6.138109	-1.712114
Malawi	0.0018666667	-9.251859	-1.775555
Guyana	0.0034666667	-17.44679	-2.722718
Guinea	0.0066666667	-19.18748	-2.482119
Cambodia	0.0088000000	-71.79817	-8.050867
Pakistan	0.0100000000	-1150.168	-100.5499
Mali	0.0174666667	-19.80574	-0.8674726
Sudan	0.0221333333	-138.8830	-1.688069

Below we inspect a few extreme examples.

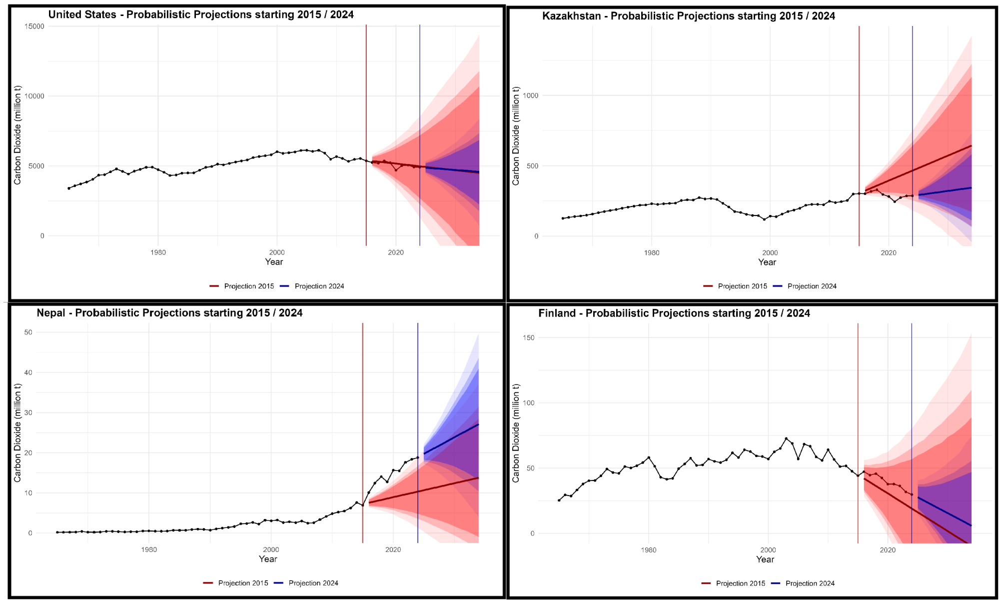

Figure 3: Probabilistic Projections of four countries

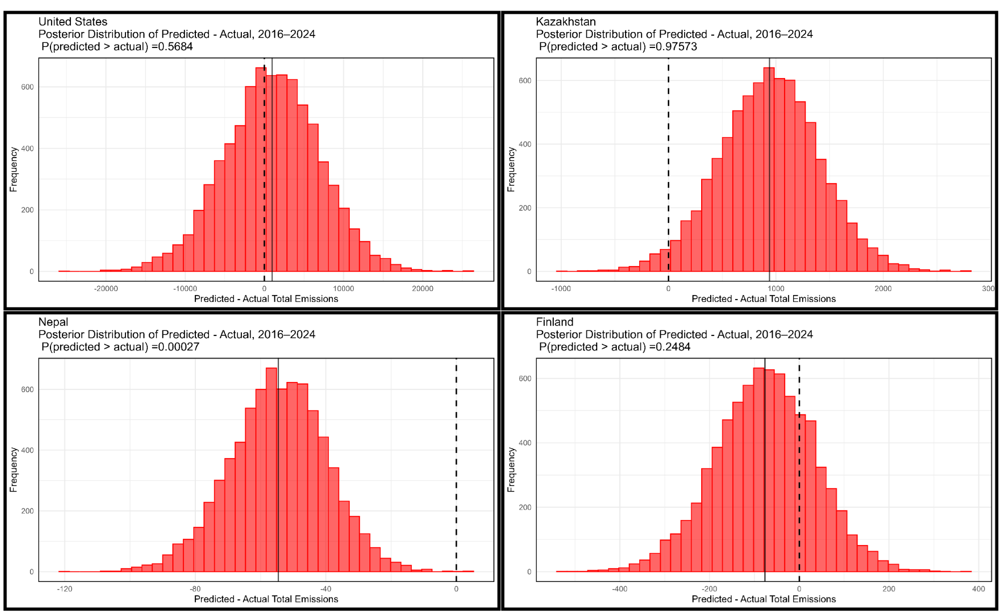

Figure 4: Posterior Distribution Difference of four countries

In the *Probabilistic Projections* figure, the forecasted and observed values for Nepal and Kazakhstan differ significantly, which explains why both countries reject the null hypothesis. An interesting case is Finland: although it reduced CO2 emissions faster than the other countries shown, it appears redder on the map. This is because Finland - like many other European countries, had already been implementing emissions-reduction policies before the Paris Agreement; consequently, a pre-2015 forecast would already have expected continued declines. So to be clear, that means we are comparing country's performance on reducing CO2 emissions before 2015 and after 2015.

5. Conclusion

In conclusion, Most countries show no statistically significant change from their pre-2015 emission trends. A small number of countries, such as South Sudan, Mozambique, Kazakhstan, etc., exhibit clearly lower emissions than predicted. On the other side we have countries such as Nepal, Cambodia, Pakistan, etc. However these results cannot be directly attributed to the Paris Agreement without further causal analysis. Overall, in a future it will be better to use more data for better evidences. For instance, using GDP, population, carbon emission per kWh of energy, and etc. would give us more information about this topic.

6. Appendix

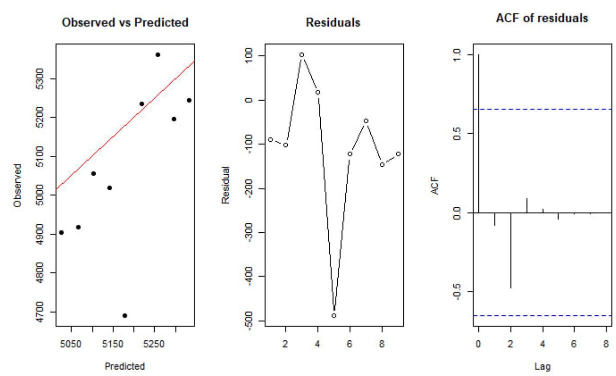

Figure 5: Residuals, Autocorrelation for United States, alpha=0.1

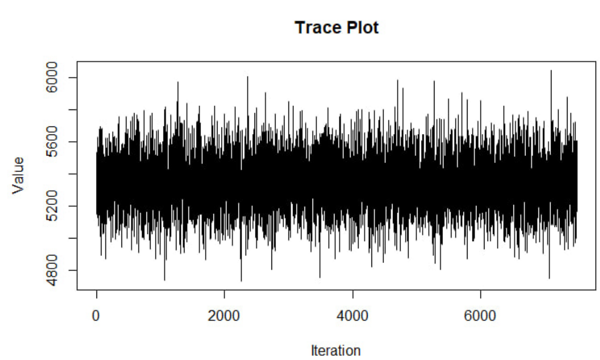

Figure 6: Trace Plot for United States, alpha=0.1